

Credit-Card Default Prediction

Casadei Alessandro — s349386
DSML Lab Project Summer 2026

Abstract—This report describes a reproducible tabular classification pipeline for predicting next-month credit-card default. The final system uses cleaned categorical variables, domain-driven repayment and billing features, a CatBoost classifier, stratified validation, and Macro F1 threshold tuning. The selected setup is the CatBoost run with seed 123 and a fixed threshold of 0.328. Archived out-of-fold (OOF) validation reached Macro F1 0.7114 at the OOF-selected threshold, while the OOF probability scores obtained ROC-AUC 0.7893. The fixed submission threshold gave OOF Macro F1 0.7106 and public leaderboard Macro F1 0.721.

I. PROBLEM OVERVIEW

The task is binary credit-card default prediction. For each client, the model predicts whether `default.payment.next.month` equals 1. The official metric is Macro F1, defined here as the unweighted average of the class-wise F1 scores, so the minority default class must influence model selection even though it represents only 22.1% of the development data.

The development set contains 24,000 rows and 25 columns, including the target. The evaluation set contains 6,000 rows and 24 columns. The feature schema is consistent with the credit-card default benchmark described by Yeh and Lien [1]. No external data were used. Table I summarizes the main data checks and exploratory findings used to guide feature engineering, while Fig. 1 shows the class imbalance and the observed effect of the most recent repayment-delay status.

The exploratory analysis showed several non-linear and asymmetric effects. Credit limit was strongly related to default risk, bill amounts were highly skewed, and adjacent monthly bill variables were highly correlated. Zero and negative bill amounts occurred in the data and were treated as informative financial states rather than errors. The EDUCATION and MARRIAGE variables contained nonstandard categories, and age risk was not monotonic: both younger and older groups had higher default rates than the central age bands.

TABLE I
DATASET AND EDA SUMMARY

Item	Value
Development shape	24,000 rows, 25 cols.
Evaluation shape	6,000 rows, 24 cols.
Default distribution	5,309 / 24,000, 22.1%
Duplicate rows / duplicate IDs	0 / 0
Development/evaluation ID overlap	0
Missing values in expected columns	0
Lowest vs. highest LIMIT_BAL bin default rate	32.2% vs. 12.7%
Recent delay default rate, delayed vs. not delayed	50.4% vs. 13.7%

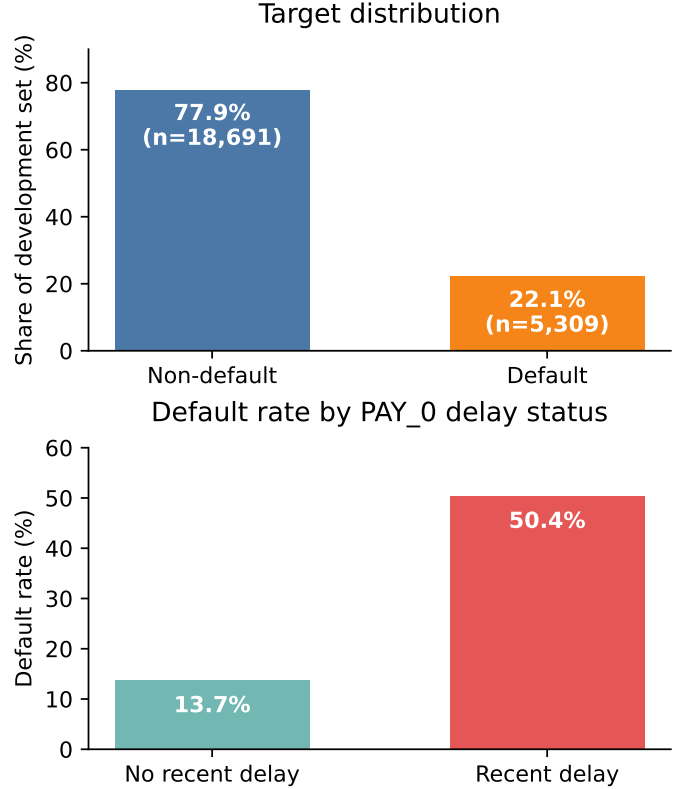


Fig. 1. Compact development-set EDA: target distribution and default rate by most recent repayment-delay status.

For example, `BILL_AMT1` had median 22,330 but maximum 964,511, with skewness 2.69. Adjacent bill months were also close substitutes (`BILL_AMT1/BILL_AMT2` Pearson correlation 0.952), so the model needed both raw monthly values and compressed summaries. Age bins further confirmed the non-monotonic pattern: the youngest and oldest bins had default rates around 25.1% and 25.2%, compared with 19.3% in the 30–34 bin.

II. PROPOSED APPROACH

The pipeline loads the development and evaluation files, detects the ID column, and removes ID from the predictive feature matrix. The target is used only for training and validation. Before one-hot expansion, the final feature table contains 65 columns: 23 original predictors, excluding ID and target, and 42 engineered features.

A. Data Integrity Controls

Before modeling, the implementation checks the parts of the pipeline most likely to create accidental leakage. The ID column is excluded from predictive feature groups, the target is present only in the development file, evaluation labels are never loaded, and no external datasets are used. After feature engineering, development and evaluation matrices are checked for identical column order, unchanged row counts, absence of target and ID columns, and absence of infinite values. These checks matter because each fold-trained model produces evaluation-set probabilities that are later averaged; even a small feature-schema mismatch would silently corrupt the final evaluation probabilities.

B. Preprocessing

Categorical cleaning is applied before feature engineering. Invalid EDUCATION codes are grouped into category 4, and invalid MARRIAGE codes are grouped into category 3. The repayment-status variables are retained because positive values directly encode payment delays. Expected numeric columns contain no missing values, but the preprocessing pipeline still uses median imputation for numeric variables and most-frequent imputation for categorical variables to keep the reproduction robust. In the final configuration, categorical variables are one-hot encoded with unknown categories ignored.

C. Feature Engineering

Feature engineering focuses on credit-risk behavior observed across the six monthly repayment, bill, and payment sequences. The useful engineered groups were repayment delay summaries; recent delay indicators; positive-delay summaries; zero and negative bill-state indicators; bill and payment summaries; payment-to-limit ratios; payment-to-bill ratios; and bill, payment, and delay trend features. These groups are meaningful for credit risk because repayment delays directly measure delinquency, recent delays emphasize the current financial state, bill summaries approximate exposure, and payment ratios measure repayment capacity relative to balances and credit limits. Trend features capture whether risk is improving or worsening across the six-month history. In the final configuration, zero-bill counts were retained, while explicit negative-bill flags were reviewed during feature engineering but pruned from the selected setup. Ratio features are clipped to reduce the influence of extreme denominators while preserving the information in zero and negative financial states.

D. Model Selection

Baseline and tree-based approaches were considered, including simple baselines, linear models, random forests, gradient boosting, LightGBM, XGBoost, and CatBoost. CatBoost was selected because boosted decision trees perform well on heterogeneous tabular data and can model non-linear interactions effectively; the algorithm is also widely used for strong tabular learning [3]. In this project, CatBoost showed the strongest validation and public transfer among the tried

TABLE II
ENABLED FEATURE GROUPS IN THE FINAL SETUP

Group	Examples	Credit-risk rationale
Delay summaries	max, mean, std delay	Captures severity and persistence of delinquency.
Recent delay flags	PAY_0-based indicators	Emphasizes the most current repayment state.
Positive delay counts	delayed months, positive delay sum	Separates risky positive delays from duly paid months.
Bill summaries	mean, max, total bill	Approximates exposure and balance pressure.
Payment summaries	mean, max, total payment	Measures recent repayment capacity.
Utilization ratios	bill/payment to limit	Normalizes exposure and payment behavior by credit limit.
Coverage ratios	payment to bill, monthly coverage	Measures whether payments cover accumulated bills.
Trend features	delay, bill, payment trends	Captures whether behavior is improving or worsening.

TABLE III
RETAINED MODEL COMPARISON EVIDENCE

Model / setup	Evidence	Decision
Dummy baseline	Holdout Macro F1 0.501	Sanity baseline
Logistic regression	Holdout Macro F1 0.695	Scaled linear baseline
RF / gradient boosting	Holdout Macro F1 0.704 / 0.708	Tree baselines
LightGBM	Holdout Macro F1 0.709	Boosted-tree alternative
CatBoost + LightGBM blend	OOF Macro F1 0.7103	Rejected
Five-seed ensemble	OOF Macro F1 0.7101	Rejected
CatBoost seed 123	OOF best 0.7114; public 0.721	Selected
Catboost seeds 777/999/2026	Similar OOF, weaker public transfer	Rejected
L2 variants 20 / 30	Marginal OOF changes	Rejected
Feature ablations	no trends, no coverage, no zero counts	Rejected
Stacking / outlier variants	No reliable public gain	Rejected

alternatives. The retained comparison evidence is summarized in Table III.

Several variants were explored before selecting the final run: different random seeds, L2 regularization settings, feature ablations, threshold sweeps, stacking or blending attempts, reranking ideas, and outlier-robustness variants. These more complex variants did not improve performance reliably. The final model was selected because it reached the best observed score while remaining a single, reproducible CatBoost setup.

E. Validation Protocol

The work used two validation stages. First, an 80/20 stratified holdout split with inner stratified cross-validation supported the initial model and hyperparameter search. This stage was useful for fast comparison among model families, but it did not determine the final submission by itself.

The final assessment used five-fold stratified out-of-fold validation over all 24,000 development rows. For each fold, the model was trained on the remaining folds, validation probabilities were written into the OOF vector, and evaluation-set probabilities were produced by the fold-trained model. The final evaluation probability for each client was the mean of the five fold predictions. This gives both an OOF validation estimate on all development rows and a deterministic submission procedure without using the evaluation labels.

TABLE IV
FINAL CATBOOST CONFIGURATION

Parameter	Value
Iterations	1100
Learning rate	0.02
Depth	6
L2 leaf regularization	25
Random seed	123
OOF folds	5 stratified folds
Class weighting	None at model level
Categoricals	One-hot encoded
Decision threshold	0.328

TABLE V
OOF VALIDATION AT FIXED THRESHOLD 0.328

Class	Precision	Recall	F1	Support
Non-default	0.868	0.884	0.876	18,691
Default	0.564	0.527	0.545	5,309
Macro avg.	0.716	0.706	0.711	24,000

F. Hyperparameter Tuning

Model assessment used stratified validation because the default class is minority. Hyperparameter search was restricted to the available development data and did not use external data. The CatBoost search space included the number of iterations, learning rate, tree depth, and L2 leaf regularization. The final reproducible setup uses seed 123, 1100 iterations, learning rate 0.02, depth 6, and L2 leaf regularization 25.

Class imbalance was handled at the validation and decision stages: the selection metric was Macro F1, folds were stratified, out-of-fold predictions were used for assessment, and probability thresholds were swept instead of using the default 0.5 cutoff. The selected final threshold is fixed at 0.328, which makes the submitted predictions deterministic.

III. RESULTS

The final submission was generated by the CatBoost setup with seed 123 and fixed threshold 0.328. The archived OOF threshold sweep selected 0.3215, where OOF Macro F1 was 0.711384. The OOF probability scores obtained ROC-AUC 0.789282. The fixed threshold used for the submitted file gives OOF Macro F1 0.710586. Its OOF confusion matrix at this threshold is $\begin{bmatrix} 16528 & 2163 \\ 2510 & 2799 \end{bmatrix}$, using rows as true non-default/default and columns as predicted non-default/default. Table V gives the corresponding class-wise metrics.

At threshold 0.328, the OOF validation contains 2,163 false positives and 2,510 false negatives. The former represent clients incorrectly flagged as risky, while the latter are missed defaulters and are usually the more costly error in credit-risk applications. The default-class recall is therefore only 0.527, showing that the model improves the baseline substantially but still misses a large fraction of difficult default cases.

Threshold tuning was important because Macro F1 penalizes weak performance on the minority default class. The selected threshold is below 0.5, increasing the number of predicted defaults relative to a standard probability cutoff. Nearby

TABLE VI
FINAL MODEL AND RESULT SUMMARY

Item	Result
Final model	CatBoost
Seed	123
Fixed threshold	0.328
Predicted defaults	1,258 / 6,000
Predicted non-defaults	4,742 / 6,000
OOF-selected threshold	0.3215
OOF Macro F1 / ROC-AUC	0.7114 / 0.7893
Public Macro F1	0.721

TABLE VII
NEARBY THRESHOLD SWEEP FOR THE SELECTED RUN

Threshold	OOF Macro F1	Eval defaults	Public F1
0.323	0.7107	1,279	0.718
0.326	0.7106	1,269	0.720
0.327	0.7107	1,263	0.720
0.328	0.7106	1,258	0.721
0.329	0.7105	1,253	0.719

thresholds produced similar but slightly different prediction counts, as shown in Table VII; the final threshold was fixed to keep the submitted distribution and file hash reproducible.

Reproducibility. The final configuration fixes random seed 123 and decision threshold 0.328. Running the final project code deterministically generates the root-level `submission.csv` with 6,000 rows and exactly the columns `Id` and `Predicted`. The submitted distribution is 1,258 predicted defaults and 4,742 predicted non-defaults. No external data were used.

IV. DISCUSSION

The strongest signal comes from repayment history, especially recent delay status and summaries of positive delays over the six-month sequence. Credit limit and bill/payment dynamics add complementary information: lower credit limits correspond to higher observed default rates, while payment-to-bill and payment-to-limit ratios capture repayment capacity relative to exposure.

This interpretation is supported by the development-set EDA: recent repayment delay is the strongest visible risk signal, while the lowest `LIMIT_BAL` bin shows a 32.2% default rate compared with 12.7% in the highest bin. Bill-to-limit and payment-to-bill ratios help normalize exposure and repayment capacity, and trend features capture whether recent behavior is improving or worsening.

At threshold 0.328, the OOF validation contains 2,163 false positives and 2,510 false negatives. False negatives are the more critical error in a credit-risk setting, because missed defaulters may correspond to direct financial loss. The default-class recall of 0.527 shows that the model captures many risky clients but still misses a substantial fraction of difficult cases.

The ablation results are also informative. Removing trend features reduced the OOF score around the relevant threshold region, which suggests that the direction of recent behavior contains useful information beyond the current month alone. Payment-coverage and zero-count variants sometimes looked

competitive in OOF validation, but they did not transfer reliably to the public score. This was the main reason for keeping the final feature set conservative: features were retained when they had a clear credit-risk interpretation and did not introduce unstable leaderboard behavior.

The final model is deliberately simple relative to the variants explored. More complex stacking, blending, reranking, seed, feature-ablation, and robustness experiments did not produce a reliable improvement. Keeping a single CatBoost model with explicit feature engineering and a fixed threshold reduces the risk of irreproducible leaderboard fitting.

The threshold analysis shows a narrow operating region. Nearby threshold submissions showed a narrow operating region: moving from 0.323 to 0.328 reduced the number of predicted defaults from 1,279 to 1,258 and increased the observed public Macro F1 from 0.718 to 0.721, while 0.329 already dropped to 0.719. This means that ranking quality and calibration are both relevant: the model probabilities are useful, but the final decision boundary is sensitive because Macro F1 balances the majority and minority classes equally.

The main limitations are tied to model selection feedback. Public leaderboard results can introduce selection bias, especially when threshold sweeps are checked against public scores. The public score may also differ from private or otherwise unseen test performance, so no claim is made about global optimality. The final model should be interpreted as the best reproducible configuration found in the project workflow, not as a guaranteed optimum.

Further work should therefore avoid simply submitting more nearby thresholds. More meaningful extensions would include a cleaner nested-validation protocol for threshold selection, probability calibration analysis, explicit feature-importance or attribution reporting, and cost-sensitive evaluation if the application assigns different costs to false positives and false negatives. These would make the model easier to interpret and would reduce dependence on public leaderboard feedback.

REFERENCES

- [1] I.-C. Yeh and C.-H. Lien, "The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients," *Expert Systems with Applications*, vol. 36, no. 2, pp. 2473–2480, 2009.
- [2] I.-C. Yeh, "Default of credit card clients," *UCI Machine Learning Repository*, 2009, doi: 10.24432/C55S3H.
- [3] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: Unbiased boosting with categorical features," in *Advances in Neural Information Processing Systems*, 2018.
- [4] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.